

FRITZ RJHAY L. DELA CRUZ

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Final-year Computer Engineering student specializing in applied AI and full-stack/embedded development. Ships production systems independently end-to-end — including a real-time crime-detection system running as a self-hosted live pilot on real CCTV footage.

Seeking remote roles (full-time or contract) in applied AI, full-stack, or embedded engineering.

TECHNICAL SKILLS

AI / Computer Vision: YOLOv8/v11, X3D video classification, PyTorch transfer learning, prompt engineering, Ollama & LLM API integration, n8n workflow automation

Languages: Python, TypeScript/JavaScript, C/C++, Java, C#, Rust, Dart

Web & Mobile: React, Next.js, React Native, Flutter, Tauri, HTML/CSS

Backend & Data: FastAPI, PostgreSQL, SQLite, Redis, Firebase, Supabase

Embedded & Hardware: ESP32, PIC16F84A (Assembly), AutoCAD, Fusion 360, Tinkercad

FEATURED PROJECTS

EcoVision Security Sentinel — *Real-time crime detection platform — self-deployed live pilot*

YOLOv11 · X3D-XS · FastAPI · PostgreSQL · Next.js · Electron · github.com/unknowndevice077/ecovision-crime-detection-ai

- Built and ran a live pilot processing real (non-benchmark) CCTV footage for weapon, violence, and multi-person activity detection, with human-in-the-loop alert review.
- Trained a YOLOv11 weapon detector (94.1% mAP@50, 90.3% recall) on a merged 24K-image dataset. Diagnosed and fixed a hidden defect capping the X3D-XS violence classifier at ~70% — raised held-out test accuracy to 95.0%, then validated against real street CCTV, surfacing a 44.3% false-positive rate on that source — the current focus.
- Built the whole incident-management system around it: FastAPI/PostgreSQL backend, Next.js dashboard, an Electron desktop app, multi-tenant role-based access (barangay/PNP), and an ESP32-triggered siren for real alerts.

Notion AI Assistant — *AI workflow tool built for a client*

Rust · Tauri 2 · SQLite · Ollama · github.com/unknowndevice077/notion-ai-assistant

- Built for a client to organize their work through natural-language prompts alone — the app turns a single prompt into organized pages and tasks directly in Notion, no manual setup required.
- Sole developer of the full stack: OS-keyring credential storage, local Ollama or external API support (no cloud dependency), and cross-platform (Windows + macOS) builds via automated CI/CD.

Studia — *AI-powered student assistant — defended 3rd-year software thesis*

Flutter · Firebase (Auth/Firestore) · SQLite · Firebase AI · github.com/unknowndevice077/Studia

- Built a cross-platform (Android, iOS, Web, Windows, macOS, Linux) Flutter app unifying class scheduling, course-file management, and productivity tools (Pomodoro timer, lecture recording); defended as a 3rd-year software thesis.
- Shipped AI-powered study features — an AI chatbot and automatic quiz generation from uploaded course materials, including audio-transcription-to-quiz — backed by Firebase Auth/Firestore sync and SQLite offline storage.

Aurum — *XAU/USD automated trading simulator*

Next.js · TypeScript · Upstash Redis · github.com/unknowndevice077/aurum-xauusd-auto-trader

- Built three trading modes — a persistent server-side bot, a browser-based simulator, and a historical backtester — sharing one signal engine for directly comparable results.
- Implemented EMA/RSI/MACD/Bollinger Band signals with Kelly-criterion position sizing, plus a backtest engine that reconstructs true equity curves for drawdown analysis against real historical gold price data.

Additional projects (12+ repos on GitHub): Line Following Robot, LeadPilot (job-board matching), plus embedded builds (fire alarms, toll gates, energy harvesting).

WORK EXPERIENCE

IT & Marketing Assistant (OJT) — Carlosta Hotel

Jun 2021 — Jul 2021

- Maintained hotel IT systems and network devices; designed promotional materials for marketing campaigns.

EDUCATION

Western Leyte College — Ormoc City, Philippines

Bachelor of Science in Computer Engineering (BSCpE) — In Progress

- Relevant coursework: Embedded Systems Design, Computer Vision & Object Detection, Robotics Kinematics, Microprocessor Systems (PIC16F84A Assembly), Cloud Frameworks & Database Management. Senior High (ICT strand), graduated with honors — Outstanding Awardee in Technical-Vocational Education.

COMPETITIONS

CodeChum National Programming Challenge 2024

Certificate of Participation

- Competed nationally in algorithmic problem-solving (trees, graphs, hash maps, DP, greedy).

ISITE AI Hackathon 2026

HexCorePH Labs

- Built QuestScribe, an n8n-automated gamified learning platform with an AI chatbot tutor, for the hackathon's automation brief (learning track).